SHAWN WEI

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Education

University of Waterloo *Sept 2025 – Apr 2029*

Honours Bachelor of Computer Science, Co-op Waterloo, ON

- Awarded the **Richard and Elizabeth Madter Entrance Scholarship** as the **sole recipient** from the incoming Faculty of Mathematics cohort for 2025

Experience & Leadership

TD Bank *May 2026 – Aug 2026*

Data Engineering / Marketing Analytics Toronto, ON

- Engineered distributed **PySpark** (Python + SQL) pipelines on **Azure Databricks** to clean and transform relational data across **15M+** active Canadian customer profiles
- Built a **PySpark** notebook-generation tool that templates **SQL** and Python queries against the customer database, standardizing **400+** analytics notebooks and eliminating inconsistent manual querying
- Built **cross-reference validation** tooling to reconcile records across migrated datasets, catching data leaks that would have launched marketing campaigns on **inaccurate data**
- Engineered a **TD Prism**-powered **RAG assistant** over TD's internal **Salesforce CDC** database, giving **170+ colleagues** self-serve natural-language access to **1,350+ marketing segments**; recognized by the **Chief Marketing Officer**

Wat.ai *Sept 2026 – Present*

AI Software Engineer Waterloo, ON

- Engineered a low-latency **OpenCV** vision pipeline applying dynamic perspective transforms on live camera feeds, extracting physical pen movements into sequential Stroke-5 vector data at **30 FPS** (**<50 ms** latency)
- Trained generative sequence prediction models (**TensorFlow** Transformers, MDN RNNs) to anticipate human sketches, converting neural outputs into **G-code** to drive real-time CNC hardware actuation with sub-millimeter precision

FRC Team 9569 Silver Knights Robotics *Oct 2022 – Jan 2025*

Programming Lead Markham, ON

- Orchestrated an **8-person** team's software lifecycle during a 6-week build season, architecting a modular **Java**/**WPILib** codebase that enforced Agile and **Git** strategies to reduce technical debt by **25%**
- Programmed autonomous **OpenCV** and **AprilTag** vision routines for real-time target acquisition and **PID** path planning, securing a **Rookie All-Star Award** and top-40% provincial ranking (**53/138**)

Skills

- **Languages:** Python, Java, C, C++, JavaScript, TypeScript, SQL, Bash
- **AI/ML & Data:** TensorFlow, LLM fine-tuning (LoRA), FP8 quantization, PySpark, Pandas, NumPy, Databricks
- **Frameworks:** React.js, Next.js, Node.js, FastAPI, RestAPIs, Flask, Django, TailwindCSS, WebSockets
- **Tools & Platforms:** Azure, Docker, Git, Linux/Unix, CI/CD, OpenCV, Salesforce

Projects

Ensemble – Finalist at Hack the 6ix *Jul 2026*

Python (asyncio), JavaScript, C++ | WebSockets, OpenCV, Qwen3.5 (LoRA), Fireworks, Arduino

- Demoed live to **230+ hackers** and sponsors, turning **200+ audience phones** into one gesture-conducted orchestra with **synchronized** browser audio via a **Python asyncio** + **WebSocket** server using NTP-style clock sync
- Fine-tuned **two Qwen3.5 LoRA** models that compose accompaniment live each bar, merged to **FP8** and served on a dedicated **Fireworks H200** at **453 ms/bar** ($10\times$ faster than shared serving)
- Guaranteed **gapless** playback by prefetching accompaniment **2 bars (~4.8 s)** ahead and falling back to a deterministic engine on any missed request

Real-Time Flight Telemetry & Analytics Pipeline *Dec 2025*

Python, JavaScript | Flask, SQLAlchemy, PostgreSQL, NumPy, Leaflet, OpenSky API

- Engineered a high-throughput ingestion pipeline to process real-time **ADS-B** signals, ensuring **data integrity** while transforming unstructured JSON telemetry into relational schemas
- Reduced query retrieval latency by **40%** for **50+** concurrent streams by implementing efficient **indexing strategies** and local caching
- Applied **statistical analysis** with **NumPy** to surface telemetry trends, providing actionable insights into aircraft performance metrics